Name: ______________________

Class: ______________________

Date: ______________________

Mr. Rawson • GCSE Physics

Kinematics L3: Constant Velocity — ANSWERS

Constant velocity

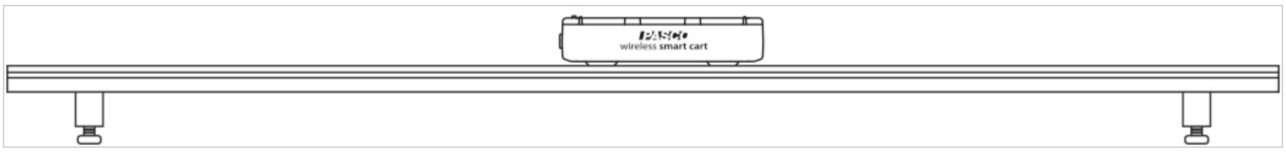
Velocity is displacement divided by the time taken:

$$\text{velocity} = \frac{\text{displacement}}{\text{time}}$$

Displacement is a vector — it carries a *direction* as well as a size. Divide a vector by a time and what you get is still a vector, so **velocity is a vector too**. That is exactly what separates velocity from speed: velocity also tells you *which way*.

So a cart moving at constant velocity is not only changing its position — it is changing position **in a particular direction**, at a steady rate. Today you record that same motion two ways at once, as a **position–time** graph and a **velocity–time** graph, and look for the pattern that connects them.

Set up



The track must be LEVEL — no books under either end. Equipment illustration courtesy of PASCO scientific.

1. Put the track flat on the bench. Adjust the feet until a cart placed in the middle **does not creep either way**. This matters — a sloping track will not give you constant velocity.
2. Press the cart's **ON** button. Check the Bluetooth light is **blinking red**.
3. On the laptop, connect the cart in SPARKvue and build a graph of **position** and a graph of **velocity**, both against time. *(See the SPARKvue guides if you are stuck.)*
4. Find the $+X$ **arrow printed on top of the cart**. That arrow is the positive direction. Point it **away from you, down the track**, and leave it that way all lesson.

What you are looking for

Key concept

A **velocity–time graph** has **velocity on the y-axis** and **time along the x-axis**. So:

- **How far the line sits from zero on the y-axis** tells you *how fast*.
- **A line below zero on the y-axis** means it is going in the **negative direction**.
- **A line with a slope of zero does *not* mean it has stopped.** A slope of zero means the **velocity is not changing** — steady speed. Only a line sitting *on* zero means stopped.

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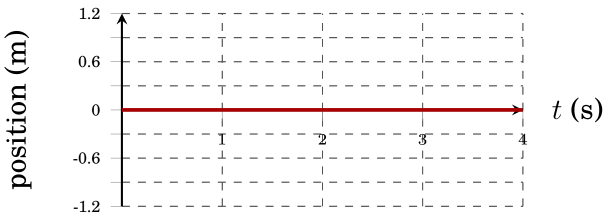
The four runs

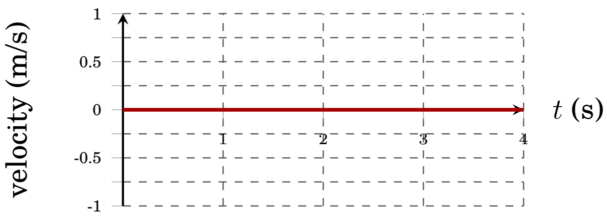
Run each one, then sketch what the screen showed. Sketch what you actually see, not what you think it ought to look like.

► **Where you start the cart changes from run to run.** Only run 1 begins in the middle. Read the start position at the top of each one — getting it wrong is what makes a graph come out the opposite way round to everybody else's. **Catch the cart at the far end every time.**

Run 1. Middle of the track. Cart sitting still, 4 seconds.

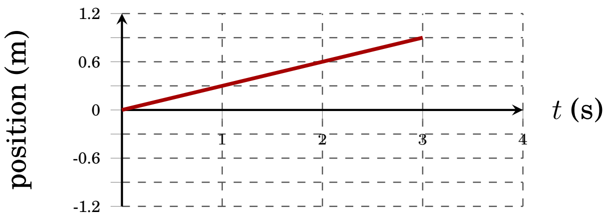
Sketch the position–time graph and the velocity–time graph.

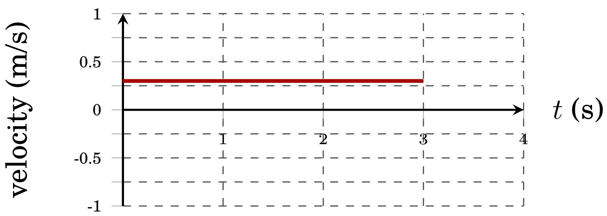




Run 2. Start at the near end. Gentle push in $+X$ — away from you.

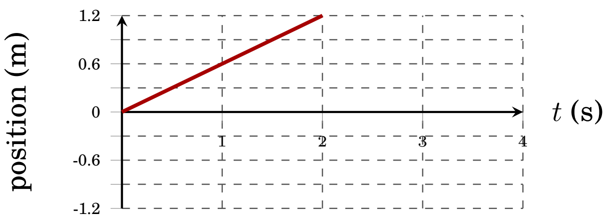
Sketch the position–time graph and the velocity–time graph.

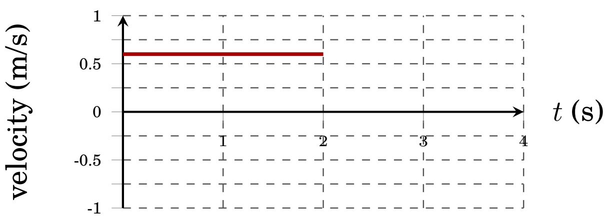




Run 3. Near end again. Harder push, same direction.

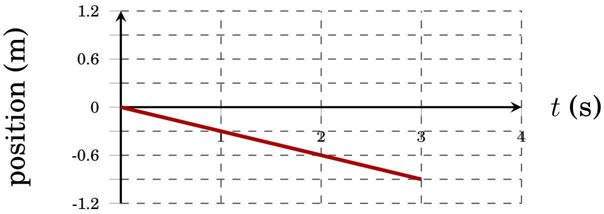
Sketch the position–time graph and the velocity–time graph.

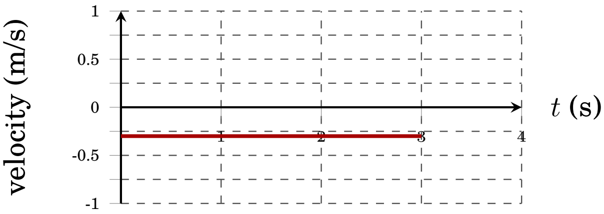




Run 4. Carry the cart to the far end. Push it back towards you — $-X$.

Sketch the position–time graph and the velocity–time graph.





Students' own traces will differ in steepness and length — the ***shape*** *and the* ***side of the zero line*** *are what is being checked. Real traces will also wobble, and will show the push at the start and the catch at the end; both are worth pointing at.*

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Your results

You are going to get the cart's velocity out of the graphs **two different ways**, and they should agree.

Key concept

SPARKvue's **Linear Fit** tool draws the best straight line through your data and gives you the equation of that line in the form

$$y = mx + b$$

where m is the **slope** and b is the **y-intercept** — the value of y where the line crosses the vertical axis.

- **On the position–time graph, m is the velocity.** Position divided by time is exactly what a velocity is, so the slope of that line *is* how fast the cart was going, with a sign telling you which way.
- **On the velocity–time graph, m should come out as zero** — the velocity is not changing — and then b **is the velocity**, because a line with a slope of zero sits at the same y value the whole way across.

Two routes, one answer. If your two numbers do not roughly match, one of the fits is wrong — most likely you have fitted the push or the catch as well as the steady part.

Highlight only the steady middle of the run before you fit. Leave out the push at the start and the catch at the end — neither is constant velocity, and both will drag your line off.

► Your velocity–time slope will almost never be exactly `0`. Write down whatever it says, even if it is something like `0.00034`. We will talk about what a number that small is telling you.

Run	Slope of x–t / $\mathrm{m\,s^{-1}}$	Slope of v–t	y-intercept of v–t / $\mathrm{m\,s^{-1}}$
1	*0*	≈ 0	≈ 0
2	*about* $+0.3$	≈ 0	*about* $+0.3$
3	*about* $+0.6$	≈ 0	*about* $+0.6$
4	*about* -0.3	≈ 0	*about* -0.3

What the shape means

1) Compare runs 2 and 3. What is different about the two velocity lines, and why?

Run 3's line also has a slope of zero, but it sits further above the zero line. A harder push gave the cart a bigger velocity, and the height of the line is the speed.

2) **Explain** what is different about run 4's velocity line, and what that tells you. Say what its *position*–time line did as well.

The velocity line has a slope of zero and sits about the same distance from zero as run 2, but it sits below the zero line — similar speed, opposite direction. Its position–time line sloped downwards, because the cart was travelling in the negative direction so its position got smaller as time went on.

3) **Explain** why a velocity–time line with a **slope of zero** does **not** mean the cart has stopped.

A slope of zero means the velocity is not changing, so the cart is going at a steady speed. Only a line lying on the zero line means it has stopped — and in runs 2 and 3 that zero-slope line sat well above zero while the cart crossed the whole track.